Three binary exceptions to a symmetry exclusion

Human Compression project: a bounded L031 correction

The supplied edition of Brooksbank–Ferrara–Leemans [1], Theorem 1.4, excludes the field of two elements from its existence statement in every dimension $d \geq 3$. There are exceptions in dimensions three, four and five. We prove them directly, including generation of the full isometry group and the intersection property. This corrects a boundary of the supplied theorem. It does not replace its positive constructions over larger fields, decide other binary dimensions, or supply a new answer to an open problem.

For $n \geq 4$ put

$$E_n = \{x \in \mathbf{F}_2^n : \textstyle\sum_i x_i = 0\}, \qquad Q(x) = \tfrac{|\operatorname{supp} x|}{2} \pmod 2.$$

The numerator is even. The identity $|A \triangle B| = |A| + |B| - 2|A \cap B|$ shows that the polar form of Q is $B(x, y) = \sum_i x_i y_i$ on E_n. Thus Q is a quadratic form. Put $v_i = e_i + e_{i+1}$, $1 \leq i < n$. These vectors form a basis, $Q(v_i) = 1$, and $B(v_i, v_j) = 1$ exactly when $|i - j| = 1$.

Lemma 1. *The symmetries $s_i : x \mapsto x + B(x, v_i)v_i$ generate a faithful copy of S_n on E_n. On this ordered generating set they form a string C-group of rank $n - 1$.*

Proof. The displayed formula interchanges coordinates i and $i + 1$, so the generators are the usual adjacent transpositions. Their action on E_n is faithful: an element acting trivially fixes every vector $e_a + e_b$, hence every two-element subset. For distinct a, b, c, the intersection of $\{a, b\}$ and $\{a, c\}$ is $\{a\}$, so it fixes every coordinate. The adjacent transpositions generate all permutations, as repeated adjacent interchanges move any entry to any prescribed position.

For a subset $I \subseteq \{1, \ldots, n - 1\}$, join i to $i + 1$ when $i \in I$. Its components are intervals. The subgroup $\langle s_i : i \in I\rangle$ is exactly the group of all permutations preserving each component: each interval has its adjacent transpositions, and generators on distinct components commute. A permutation lies in two such subgroups exactly when it preserves every intersection of a component of the first path with one of the second. These nonempty intersections are the components of the path with edge set $I \cap J$. Consequently

$$\langle s_i : i \in I\rangle \cap \langle s_j : j \in J\rangle = \langle s_k : k \in I \cap J\rangle.$$

Nonadjacent generators commute and adjacent ones have product of order three. This proves all string C-group requirements. □

Theorem 2. *For $n = 4, 5, 6$, the full isometry group of (E_n, Q) is the faithful S_n of Lemma 1. It is therefore a string C-group generated by symmetries of rank $d = 3, 4, 5$, respectively. Its polar radical has dimension one for odd d and zero for even d.*

Proof. The orthogonal complement of E_n in $\mathbf{F}_2^n$ is spanned by the all-one vector z. Hence the polar radical is $\langle z\rangle$ when n is even, and zero when n is odd.

For $n = 4$, $Q(z) = 0$. The quotient $E_4/\langle z\rangle$ has dimension two. Every isometry fixes z and induces an invertible map on that quotient, giving at most $|\mathrm{GL}_2(2)| = 6$ choices. The kernel consists of maps $x \mapsto x + f(x)z$, where f is linear and $f(z) = 0$. There are at most four such maps. Thus the isometry group has order at most 24. It already contains the faithful S_4 of order 24, proving equality.

For $n = 5$, the nonsingular vectors are precisely the ten vectors of weight two; the vectors of weight four are singular. Two distinct nonsingular vectors have polar product one exactly when the corresponding pairs share a coordinate. Every isometry therefore acts on the graph of the two-element subsets of a five-element set, adjacent when they intersect. The cliques of size four in this graph are exactly the five stars of all pairs containing one fixed coordinate. Indeed, three pairwise intersecting pairs without a common coordinate are the three edges of a triangle, and no fourth distinct pair can meet all three. Graph automorphisms consequently permute these five stars. Their action on the stars is faithful because each vertex is the unique common member of two stars. The isometry action on this graph is faithful too, since its weight-two vectors span E_5. Thus the isometry group has order at most 5!, attained by the coordinate permutations.

For $n = 6$, $Q(z) = 1$. The quotient $W = E_6/\langle z \rangle$ is a nondegenerate symplectic space of dimension four. The induced homomorphism from the isometry group into $\mathrm{Sp}(W)$ is injective. An element in its kernel is $x \mapsto x + f(x)z$ with $f(z) = 0$, and preservation of Q gives $Q(x + f(x)z) = Q(x) + f(x) = Q(x)$, so $f = 0$. To count the symplectic group, choose the first vector of a symplectic basis in 15 ways, its partner in 8 ways, and a symplectic basis of their two-dimensional perpendicular complement in $3 \cdot 2$ ways. A symplectic map is uniquely determined by the image of a fixed symplectic basis, so its order is $15 \cdot 8 \cdot 3 \cdot 2 = 720$. The faithful coordinate S_6 already has this order. Equality follows once more. □

Relation to the printed matrices. In the basis $v_1, \ldots, v_{n-1}$ the quadratic form has diagonal entries one and consecutive upper entries one, with all other entries zero. This is precisely the source's $\Phi(1, \ldots, 1)$. The group it generates is S_{d+1}; the supplied proof on physical page 8 instead writes S_d. The three full-isometry examples above disprove the stated binary exclusion, independently of how that subscript arose. The first example has singular radical, the third nonsingular radical; both are allowed by the wording of Theorem 1.4.

The proof uses only elementary linear and permutation group arguments written here. No finite program, external group-order table or formal verification is a premise. The standard permutation model and the known small-group identifications are not claimed as new ideas. The earlier authors retain attribution for the larger-field construction; its complete recovery remains a separate obligation.

References

[1] P. A. Brooksbank, J. T. Ferrara and D. Leemans, *Orthogonal groups in characteristic 2 acting on polytopes of high rank*, arXiv:1709.02219v2, 10 March 2018. The fixed supplied edition has twelve pages.